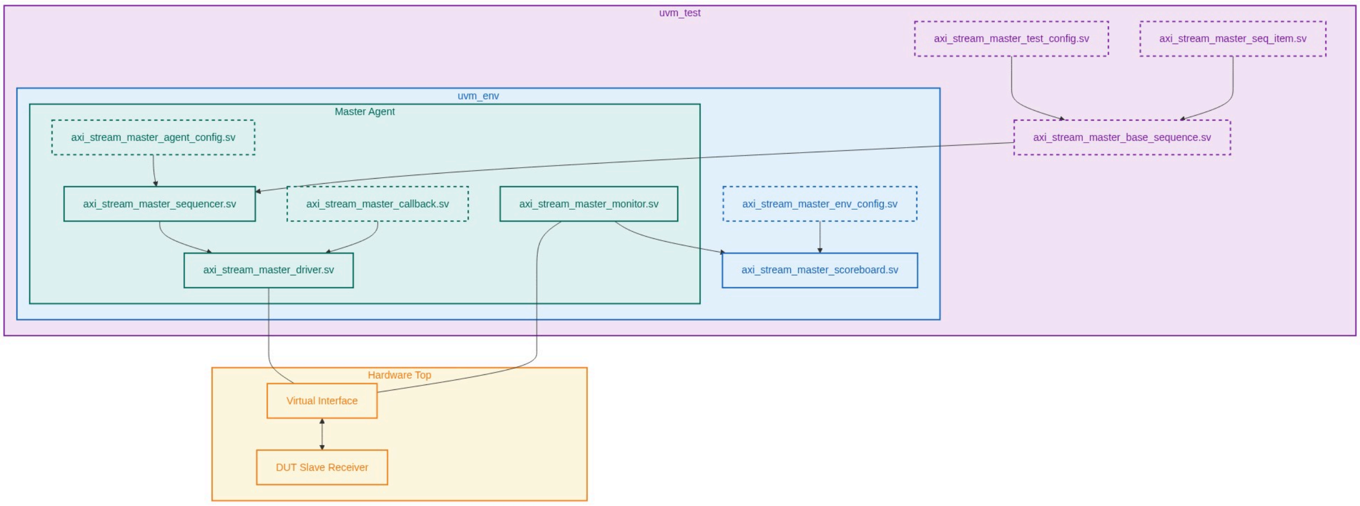
Verification Plan Report

Project: AXI-Stream Master VIP Verification Plan

Version: 14.0

Verification Role: Master

1. UVM Environment Architecture



2. Protocol Configurations

Feature	Configuration Name	Impact
Handshake Protocol	TVALID_TREADY_Async_Handshake	Functionality impact: Forces pipeline stalling and dictates stability bounds for the entire payload vector. Performance impact: dictates maximum theoretical throughput in scenarios involving combinatorial TREADY backpressure.
Packet Framing	TLAST_Packet_Framing_Support	Functionality impact: Requires end-to-end exact reconstruction of packet boundary assertions to govern data routing layer and destination buffers. Performance impact: dictates fragmentation penalties and dynamic arbitration thresholds.
Byte-Level Stream Semantics	TKEEP_TSTRB_Qualifier_Encoding	Functionality impact: necessitates robust reconstruction algorithms in the scoreboard to extract physical data while discarding positional or null bytes. Performance impact: alters actual byte-consumption rate compared to clock-cycle metrics.

Reset Sequencing	ARESETn_Mid_Transfer_Isolation	Functionality impact: compels the testing environment to trigger state-machine destruction mid-transfer. Verification constraint: requires the VIP driver to safely discard tracking state and the monitor to correctly sequence clean IDLE resets asynchronously.
Wakeup Signaling (AXI5)	TWAKEUP_Pre_Transfer_Signaling	Functionality impact: introduces an external deadlock dependency if TWAKEUP assertion lags TVALID assertion. Performance impact: latency models must explicitly adjust for TWAKEUP lead times before TREADY is physically accessible.
Stream Identification	TID_TDEST_Interleaving_Routing	Functionality impact: forces the scoreboard multiplexer logic to concurrently maintain dynamically allocated data streams to track ordering compliance. Performance impact: tests the DUT buffer saturation across varying interleave depths.
Parity Protection (AXI5)	Odd_Parity_Check_Generation	Functionality impact: requires single-bit error injection capabilities within the VIP transmission chain to exercise the DUT standard error logic. Verification constraint: the monitor must passively track hardware parity matrices parallel to the native data payload across all TKEEP permutations.

3. Test Requirements

Index	Feature/Sub	Name	Description
REQ_MST_01	Basic Operations (Handshake Synchronization)	TVALID-to-Payload Stability Dependency	Direct signal dependency: When TVALID=1 and TREADY=0, the state of TDATA, TSTRB, TKEEP, TLAST, TID, TDEST, and TUSER must not change. The protocol mandates that a Transmitter must unconditionally hold payload signals stable until TREADY=1 is sampled coincidently with TVALID=1 to prevent race conditions during sampling. Verification Intent: Ensure the VIP sequences drive stable payload signatures across highly randomized TREADY stall windows. The protocol monitor must fire fatal assertions if any payload bit changes mid-stall. This enforces strict spec compliance at the conformance boundary. Directed scenarios force standard compliance checks under various stall densities.

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REQ_MST_02	Basic Operations (Packet Boundary Tracking)	TLAST to TVALID Synchronization Constraint	Dependency Model: TLAST acts as the packet termination qualifier and holds logical meaning ONLY when TVALID=1. TLAST assertion cannot occur independent of TVALID. A packet starts immediately after a previous TLAST assertion, or on the first TVALID=1 post-reset for a given TID/TDEST tuple. Verification Intent: The VIP sequences will manipulate the interval between TVALID events and TLAST events, creating randomized multi-beat packet formations up to 4096 beats, single-beat packets, and zero-data null packets.
REQ_MST_03	Data Encodings (Byte Qualifiers)	TKEEP to TSTRB Mask Constraint	Signal Relationship: TKEEP and TSTRB share a strict dependency. TKEEP=0 designates a null byte, making TSTRB mathematically irrelevant (protocol reserved). TKEEP=1/TSTRB=1 is an evaluated data byte; TKEEP=1/TSTRB=0 is a position byte. Verification Intent: The VIP stimulus will enforce valid TKEEP/TSTRB combinations by generating sparse offsets dynamically. Negative injection handles driving the illegal TKEEP=0/TSTRB=1 condition to ensure the standard DUT error vectors trigger without modifications.
REQ_MST_04	AXI5 Features (Power Domain Wakeup)	TWAKEUP to TVALID Lead Time	Signal dependency: TWAKEUP is an AXI5 preemptive signal. If TWAKEUP transitions HIGH concurrently with TVALID, the spec dictates that TWAKEUP MUST remain HIGH until TREADY=1 completes the handshake. If TWAKEUP leads TVALID, it provides power-activation buffering logic to the Slave. Verification Intent: VIP constraints will sweep the temporal relationship: TWAKEUP asserted randomized cycles (1..32) prior to TVALID, and TWAKEUP simultaneous with TVALID. The DUT must correctly un-stall its TREADY pathway in standard operating conditions.
REQ_MST_05	AXI5 Features (Parity Error Tracking)	TDATACHK Null TKEEP Parity Constraint	Intersection analysis: TDATACHK represents the odd parity vector of the TDATA payload. The spec strictly mandates that the parity check MUST be driven across ALL physical byte lanes regardless of the TKEEP mask state. A TKEEP=0 lane must still maintain active TDATA parity protection. Verification Intent: The VIP transmitter passively calculates parity bits on all generated physical bus vectors independently of the TKEEP/TSTRB configuration. Tests will randomize data underlying disabled TKEEP bytes to verify the DUT parity error check logic.

REQ_MST_06	Exception Handling (Asynchronous Reset Isolation)	ARESETn Suppression of TVALID Vectors	Global Override: ARESETn=LOW overrides all normal operations. TVALID must remain LOW while ARESETn is LOW. TVALID must only transition HIGH on a rising ACLK edge occurring strictly after the ACLK edge where ARESETn transitions HIGH. Verification Intent: VIP forces asynchronous reset generation at peak bus congestion (TVALID=1, TREADY=0). The VIP automatically clamps TVALID=0. Monitors verify zero combinatorial bleed-through on TVALID after reset injection.
REQ_MST_07	Advanced Routing (Multi-Stream Multiplexing)	Arbitration Independence of TID Paths	Stream interdependency check: TDATA payload transfers sharing matching TID/TDEST identifiers must maintain strict temporal ordering. Transfers with distinct TID identifiers hold zero temporal interdependency and may be interleaved at transfer granularity. Verification Intent: The VIP runs a heavily randomized sequencer interleaving unique TID streams across a shared physical driver. The scoreboard passively demultiplexes TIDs and checks order-invariant progression within each unique identifier.

4. Test Cases & Mapping

Index	Scenario	TR Mapping
TC_MST_001	Directed Handshake Stalling: The VIP generator issues a directed 256-beat incremental stream. Randomization logic dynamically injects standard 0-to-4 cycle stalls.	REQ_MST_01
TC_MST_002	Directed Extreme Stall Injection: The VIP generates valid stimulus but forces randomized 50+ cycle stall gaps between TREADY loops to pressure stability.	REQ_MST_01
TC_MST_003	Directed Combinatorial Backpressure: VIP drives back-to-back continuous payloads. Randomization is applied to the DUT's simulated TREADY latency to generate 0-wait continuous flows.	REQ_MST_01
TC_MST_004	Negative Parameterized Drop: VIP is instructed to deliberately withdraw TVALID before TREADY assertion. The standard DUT behavior is observed and verified against spec compliance.	REQ_MST_01
TC_MST_005	Directed Single-Beat Packets: VIP constrained to constantly sequence 1-beat packets where TVALID and TLAST coincide on every transfer dynamically.	REQ_MST_02

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TC_MST_006	Directed Jumbo Framing: VIP generator randomized to build 4096+ beat continuous packet structures verifying boundary counters.	REQ_MST_02
TC_MST_007	Directed Null Packet Generation: VIP randomly configured to sequence Null-Size (TKEEP=all zeros, TLAST=1, TVALID=1) bursts mapping empty extraction.	REQ_MST_02
TC_MST_008	Back-to-Back Packet Sequence: VIP sequence drives a continuous payload where TLAST=1 is immediately followed by a new packet with TVALID=1 in the very next cycle without gap.	REQ_MST_02
TC_MST_009	Directed Sparse Stream Position Sequence: VIP drives randomized TKEEP variations covering fully null packages, position bytes, and mixed data streams dynamically across 500 transfers.	REQ_MST_03
TC_MST_010	Directed Full Byte Alignment: VIP forces TKEEP=1/TSTRB=1 uniformly across all physical lanes across randomized burst volumes to confirm peak efficiency processing.	REQ_MST_03
TC_MST_011	Directed Position Buffer Validation: VIP generates continuous payload where data bytes are deliberately spaced by TSTRB=0 position placeholders to test DUT reconstitution.	REQ_MST_03
TC_MST_012	Negative TSTRB Qualifier Corruption: VIP is directed to force an illegal TKEEP=0, TSTRB=1 array mask locally. VIP fault injection completes safely. Standard DUT reporting thresholds capture violation.	REQ_MST_03
TC_MST_013	Directed Simultaneous Wakeup: VIP constrained to generate TWAKEUP and TVALID simultaneously, holding both HIGH through dynamic wait periods until TREADY compliance.	REQ_MST_04
TC_MST_014	Directed Recommended Latency Sweep: VIP sequences TWAKEUP generation ahead of TVALID randomly partitioned between 1 and 4 ACLK cycles dynamically.	REQ_MST_04
TC_MST_015	Directed Extreme Pre-Activation: VIP sequences TWAKEUP 30+ cycles prior to TVALID generation to test power-domain buffering logic saturation inside the standard DUT.	REQ_MST_04
TC_MST_016	Directed Checksum Baseline Protocol: VIP calculates compliant odd parity continuously over randomized payload traffic while observing standard DUT operational throughput.	REQ_MST_05
TC_MST_017	Directed Checksum Null-Lane Sweep: VIP directed to transmit transfers where TKEEP=0 over 50% of byte lanes. VIP calculates valid parity using suppressed underlying data vectors.	REQ_MST_05

TC_MST_018	Negative TDATACHK Parity Injection: VIP explicitly flips the checksum parity bit on TDATACHK[2] dynamically during an active structural transmission. Standard DUT path resolves error.	REQ_MST_05
TC_MST_019	Negative Control Parity Injection: VIP explicitly inverts TVALIDCHK and TLASTCHK parity models randomly against standard DUT constraints.	REQ_MST_05
TC_MST_020	Directed Synchronization Reset: VIP forces ARESETn logic synchronously during completely IDLE interface sequences to guarantee non-bleeding initializations.	REQ_MST_06
TC_MST_021	Directed Mid-Packet Destruction: While VIP holds TVALID=1/TREADY=0, the VIP synchronously drives ARESETn LOW. Confirm standard DUT reliably drops pipeline cleanly.	REQ_MST_06
TC_MST_022	Directed Turnaround Injection: VIP drops ARESETn precisely coincident with a TLAST=1 valid packet culmination, destroying subsequent interlocked cycles.	REQ_MST_06
TC_MST_023	Directed Pre-Wakeup Reset Isolation: VIP asserts TWAKEUP but drives ARESETn actively low prior to TVALID transmission to verify asynchronous decoupling.	REQ_MST_06
TC_MST_024	Directed Dual-Stream Arbitration: VIP sequentially runs two separate TID threads, crossing segments cleanly without randomizing temporal logic.	REQ_MST_07
TC_MST_025	Directed High-Depth Multi-Sequence: Four parallel VIP sequences drive disjoint TID packets into a shared driver. The VIP arbitrates tightly, testing DUT tracking capacity natively.	REQ_MST_07
TC_MST_026	Directed Single-Beat Fragmentation: VIP generates 8 independent TID streams but limits sequencer grants to purely single-beat fragments constantly alternating TIDs.	REQ_MST_07
TC_MST_027	Directed Cross-Destination Interleaving: VIP sequences maintain matching TIDs but dynamically modulate TDEST headers, forcing distinct structural routing layers passively.	REQ_MST_07

5. Functional Coverage Groups

Group Name	**Description**
cg_tvalid_tready_temporal_cross	Multi-dimensional coverage cross-referencing TVALID lead times with TREADY hold times across 5 distinct duration bins.
cg_packet_boundary_dimensions	Coverage bins for packet sizes dynamically constrained by the VIP sequences (single, multi, continuous).

cg_tkeep_tstrb_combinatorial	Coverage metric tracking standard bitwise legal interactions against negative fault injections.
cg_twakeup_lead_bins	Array of tracked intervals validating coverage across all possible pre-transfer timing limits specified in testing.
cg_tdatachk_masked_integrity	Parity sequence coverage cross-referencing TDATACHK logic states mapped explicitly against TKEEP=0 vector elements.
cg_aresetn_asynchronous_window	State-machine phase bins where the reset logic was forcefully injected by the VIP.
cg_tid_fragmentation_depth	Crosses interleaved Stream IDs with maximum consecutive beats, confirming extreme edge permutation boundary interleaving.

6. Interface Signals

Signal Name	Direction	Width	Description
ACLK	Input (to VIP Master and DUT Slave)	1	Global synchronization clock.
ARESETn	Input (to VIP Master and DUT Slave)	1	Active-LOW global reset.
TVALID	Output (VIP Master / Transmitter)	1	Driven by the VIP Master. Indicates a valid transfer payload is present.
TREADY	Input (from DUT Slave / Receiver)	1	Driven by the DUT Slave to signal readiness to accept a transfer.
TDATA	Output (VIP Master / Transmitter)	TDATA_WIDTH	Primary data payload.
TSTRB	Output (VIP Master / Transmitter)	TDATA_WIDTH /8	Per-byte data qualifier.
TKEEP	Output (VIP Master / Transmitter)	TDATA_WIDTH /8	Per-byte position qualifier.

TLAST	Output (VIP Master / Transmitter)	1	Packet-boundary indicator.
TID	Output (VIP Master / Transmitter)	TID_WIDTH	Stream identifier.
TDEST	Output (VIP Master / Transmitter)	TDEST_WIDTH	Routing identifier.
TVALIDCHK	Output (VIP Master / Transmitter, AXI5-Stream only)	1	Odd parity check bit for TVALID.
TDATACHK	Output (VIP Master / Transmitter, AXI5-Stream only)	TDATA_WIDTH /8	Odd parity check per byte lane of TDATA.
TLASTCHK	Output (VIP Master / Transmitter, AXI5-Stream only)	1	Odd parity check bit for TLAST.
TREADYCHK	Input (from DUT Slave / Receiver, AXI5-Stream only)	1	Odd parity check bit for TREADY, generated by the DUT Slave.

7. Verification Environment Structure

`axi_stream_master_vip_tb/`

```
|-- master_agent/
|   |-- axi_stream_master_driver.sv
|   |-- axi_stream_master_monitor.sv
|   |-- axi_stream_master_sequencer.sv
|   |-- axi_stream_master_agent_config.sv
```

```
|   \-- axi_stream_master_callback.sv
|-- env/
|   |-- axi_stream_master_scoreboard.sv
|   |-- axi_stream_master_env_config.sv
|   \-- axi_stream_master_protocol_checker.sv
|-- sequences/
|   |-- axi_stream_master_seq_item.sv
|   |-- axi_stream_master_base_sequence.sv
|   |-- axi_stream_master_handshake_seq.sv
|   \-- axi_stream_master_interleave_seq.sv
\-- top/
    |-- axi_stream_master_tb_top.sv
    \-- axi_stream_master_test_config.sv
```